Research Statement

Zheng Ma
Tenure-Track Associate Professor of Mathematics, Shanghai Jiao Tong University
zhengma@sjtu.edu.cn | zheng-home.netlify.app | github.com/mazhengcn

1 Introduction and Research Vision

We stand at a remarkable inflection point in computational science. The most pressing open problems of our time—controlled nuclear fusion (spanning both inertial and magnetic confinement), precision subsurface imaging via seismic full-waveform inversion, accurate dose calculation in radiation therapy, and complex multiscale rarefied gas dynamics—share a common mathematical core: they require solving kinetic equations and associated inverse problems at scales, speeds, and levels of physical fidelity that are fundamentally inaccessible to classical numerical methods.

Kinetic theory provides the most complete description of matter at the mesoscale, the regime bridging quantum mechanics and macroscopic fluid dynamics. The Boltzmann equation, the radiative transfer equation (RTE), and the Vlasov–Maxwell system underpin models of plasma confinement in tokamaks, photon transport in stellar atmospheres and radiation oncology planning, and carrier dynamics in semiconductor devices. These equations define distributions over phase space of dimension $2d$ ($d = 3$ in three spatial dimensions), rendering direct discretization intractable for realistic problems. Compounding the curse of dimensionality is the ubiquitous presence of multiple temporal and spatial scales—from the mean free path of a single particle collision to macroscopic fluid motion—that force classical solvers to use extremely fine computational grids even when the macroscopic solution is globally smooth.

Deep learning has demonstrated a striking capacity to approximate high-dimensional functions and solution operators, often achieving sample complexities that defy classical worst-case predictions. Yet, the naïve application of neural networks to physical systems is plagued by fundamental limitations: black-box models routinely violate conservation laws, fail catastrophically in stiff multiscale regimes (where the Knudsen number $\varepsilon \ll 1$), and lack theoretical guarantees for out-of-distribution generalization.

Currently, the dominant trend in AI for Science mirrors that of large language models (LLMs): scaling up to massive foundation models in pursuit of a "one-model-solves-all" paradigm for PDEs. My philosophy fundamentally diverges from this monolithic approach. Instead, I advocate for developing compact, physically infused models tailored to specific governing equations—for instance, dedicating one specialized architecture to the radiative transfer equation and another to the Boltzmann equation. Within this framework, an LLM acts not as a universal PDE solver, but as an intelligent orchestrator coordinating an ensemble of domain-specific, physics-aware solvers. I term this paradigm the *Agent of Real Physical Science*.

My research program brings this vision to life through a framework I call *full-stack AI for Science*: a vertically integrated methodology comprising (i) rigorous mathematical analysis of both the physical system and the learning algorithm; (ii) neural architectures that embed physical principles as *hard constraints* rather than soft regularization penalties; (iii) scalable algorithms designed for modern accelerated hardware; and (iv) production-quality open-source software. To date, this integrated approach has yielded over 30 peer-reviewed publications and a suite of widely adopted open-source packages (hosted at `github.com/mazhengcn`) that are actively utilized across applied mathematics, physics, and engineering.

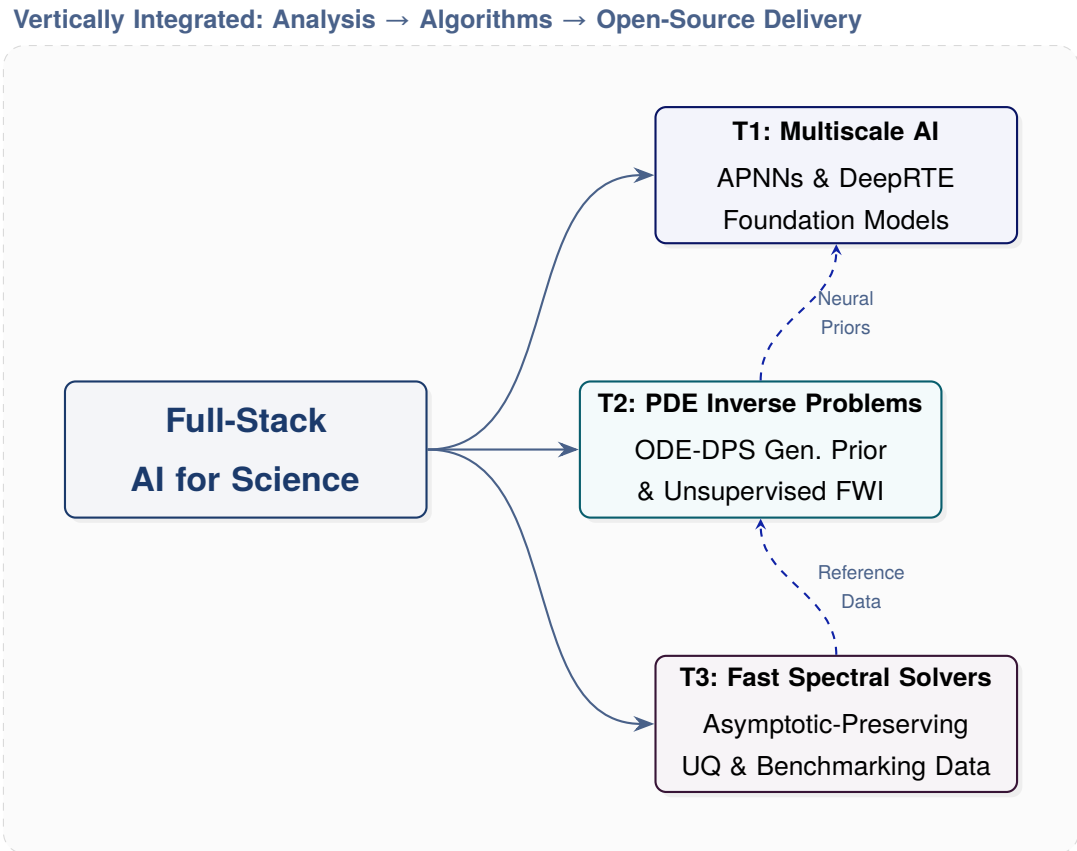

Figure 1: Overview of my research program. The unifying methodology involves rigorous mathematical analysis acting as hard constraints on AI architectures, supported by fast classical solvers for theoretical and numerical validation.

The work spans three interconnected thrusts: **(1) structure-preserving AI for multiscale kinetic equations**; **(2) AI-driven methods for PDE-constrained inverse problems**; and **(3) fast spectral algorithms** that serve as both the theoretical bedrock and the benchmark gold standard for the AI-based methods.

2 Foundation Models & Structure-Preserving AI for Kinetic Equations

Why it matters. Kinetic equations are among the most computationally expensive models in all of science. A single RTE simulation in nuclear reactor transport involves 10^8–10^{12} phase-space unknowns, with run times on classical supercomputers measured in hours.

Fusion plasma simulations face a starker challenge: real-time feedback control of a tokamak plasma requires millions of Vlasov–Maxwell evaluations per second, a throughput that is five to six orders of magnitude beyond what current particle-in-cell (PIC) codes can deliver. Equally important is the *multiscale* structure of these equations: in the limit $\varepsilon \to 0$ (small Knudsen number), the kinetic model degenerates to a macroscopic diffusion or fluid system, but classical numerical discretizations become stiff and impose a time-step restriction $\Delta t = O(\varepsilon)$, adding an entire order of computational difficulty to an already intractable problem.

Neural operators have emerged as a powerful paradigm for learning PDEs, offering capabilities that extend beyond traditional numerical solvers to modeling physical phenomena directly from experimental data. Concurrently, integrating physical constraints into machine learning models—popularized by architectures like PINNs and the Deep-Ritz method—has proven essential to significantly reduce the dependency on massive datasets. These twin principles drive our research program: first, developing **DeepRTE** to directly capture the complex structural mappings inherent in neural operators; and second, designing **Asymptotic-Preserving Neural Networks (APNNs)** to rigorously embed physical constraints into the training process, guaranteeing physical fidelity in multiscale regimes.

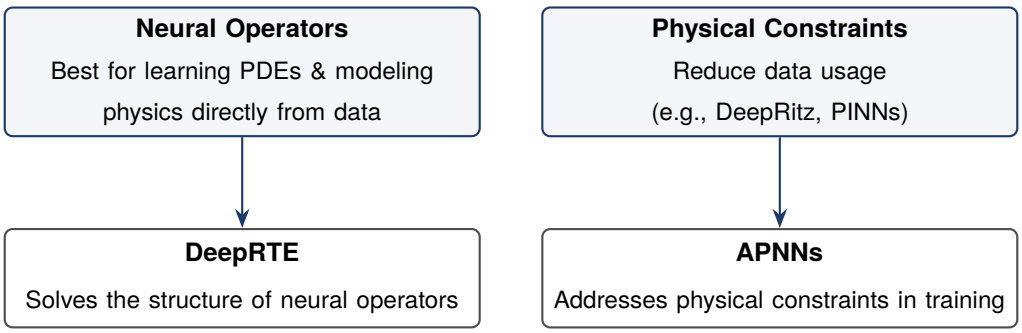

Figure 2: Two pillars of our scientific machine learning framework: data-driven operator learning and physics-constrained efficiency.

2.1 DeepRTE: A Foundation Model for Radiative Transfer

My most recent and important work is the development of **DeepRTE** [1], an efficient neural operator approach for the steady-state Radiative Transfer Equation (RTE). The RTE is a differential-integral equation that governs the propagation of radiation through a participating medium, with applications spanning diverse domains such as neutron transport, atmospheric radiative transfer, heat transfer, and optical imaging. Classical deterministic methods typically involve discretizing the equation in both spatial and angular variables, which is computationally expensive and is considered as one of the core tasks in high-performance computing. Our DeepRTE framework demonstrates superior computational efficiency for solving the steady-state RTE, surpassing traditional methods and existing neural network approaches. This efficiency is achieved by embedding physical information through derivation of the RTE and mathematically-informed

network architecture, and using a mesh-free neural operator framework with inherent zero-shot capability.

Aspect	Classical Solvers (e.g., S_N)	DeepRTE Foundation Model
Run Time (Inference)	10^4–10^6 seconds per geometry	$\sim$ 2.3 s (4× GPU, 83.9× speedup)
Mathematical Scope	Equation discretization	Direct operator mapping $\mathcal{A} : (I^-, \mu_t, \mu_s, p) \mapsto I$
Reusability	None natively	Zero-shot generalization without retraining
Distribution	Specialized Fortran/C++ libraries	Hugging Face checkpoints, JAX/Flax APIs

Figure 3: A paradigm shift in radiative transfer: DeepRTE replaces costly per-instance simulation with efficient inference across variable geometries.

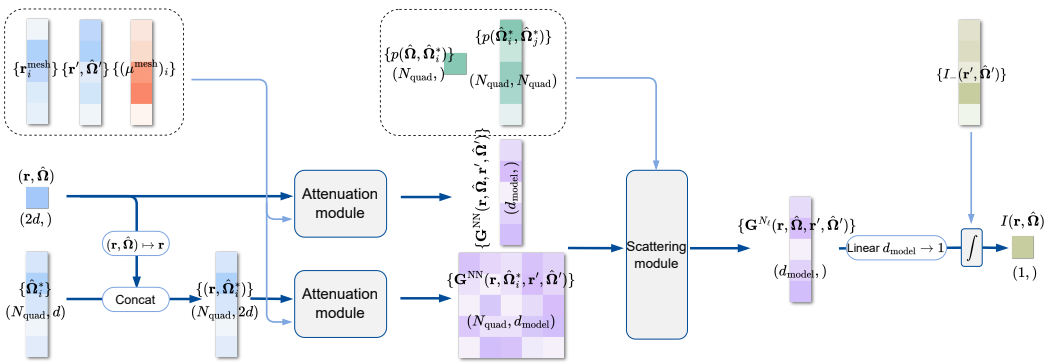

Figure 4: DeepRTE network architecture. The diagram illustrates the alterations in various inputs throughout the process. The Attenuation Module processes phase coordinates and coefficients to approximate operators $\mathcal{J}$ and $\mathcal{L}$, followed by the Scattering Module handling angular quadrature evaluations for operator iterations. Ultimately, the Green's function G^{NN} is multiplied by the boundary conditions and integrated over Γ_- to compute the radiation field.

The engineering realization is as critical as the mathematical novelty. DeepRTE is implemented entirely in JAX/Flax for maximum hardware acceleration, containerized with Docker for reproducible deployment, and distributed via:

- GitHub package with full training/inference pipeline: `github.com/mazhengcn/deeprte`
- Pre-trained checkpoints and datasets on Hugging Face: `huggingface.co/mazhengcn/deeprte`

This represents a qualitative shift in computational physics: rather than an isolated scheme for an individual problem, DeepRTE provides the community with a *reusable computational asset* that amortizes the immense initial training cost across thousands of downstream, domain-specific tasks.

2.2 Asymptotic-Preserving Neural Networks Framework

The central contribution of my group in this area is the development of the *Asymptotic-Preserving Neural Networks* (APNNs) framework. This is not merely a single model, but a fundamentally new series of physics-informed methodologies to address multiscale problems and asymptotic-preserving (AP) properties.

The key insight driving this framework is that when using neural networks to solve PDEs, the equations are inherently transformed into optimization problems. Consequently, the hard constraints of classical solutions and parameters are relaxed into soft constraints. In a data-driven setting or within pure PINN setups, such soft constraints fail drastically in extreme multiscale regimes (e.g., Knudsen number $\varepsilon \to 0$), leading to severe performance degradation due to spectral bias and stiffness.

To rectify this, APNNs decouple the multiscale constraints. We incorporate the micro–macro decomposition algebraically into the overall construction, forming multiple or even redundant equations to guide the relaxed optimization safely toward the correct limit. Essential physical properties—such as mass/density conservation or Maxwellian convergence—are inherently satisfied as **hard architectural constraints**. This formulation guarantees that even with highly limited labeled data and the presence of optimization inaccuracies, the learned network will naturally preserve uniform accuracy across all scales and correctly degenerate to the macroscopic limits. The framework is flexible and not bound merely to PINNs; it thrives seamlessly with DeepRitz or general neural-operator paradigms.

$$\begin{tikzcd}
\mathcal{R}(\mathcal{F}^\varepsilon) \arrow[r, "\varepsilon \to 0"] \arrow[d, "\mathcal{R} \to 0"] & \mathcal{R}(\mathcal{F}^0) \arrow[d, "\mathcal{R} \to 0"] \\
\mathcal{F}^\varepsilon \arrow[r, "\varepsilon \to 0"] & \mathcal{F}^0
\end{tikzcd}$$

Figure 5: Diagram illustrating the APNN framework.

The framework has successfully revolutionized the treatment of numerous canonical multiscale kinetic systems. We summarize these works below:

- **Basic APNN for Linear Transport** [2, 3]: We first established the APNN framework using the even-odd decomposition for linear transport equations. We proved uniform accuracy from the kinetic regime to the diffusion limit. The network achieves optimal performance on problems where standard PINNs fail completely for $\varepsilon < 10^{-3}$, without relying on strict mesh refinement.
- **APNN for Non-linear Systems (VPFP)** [4]: Moving to high-field regimes, we extended APNNs to the Vlasov–Poisson–Fokker–Planck system. By adapting the loss formulation and microscopic structures, the updated network captures AP behavior under complex, long-range electromagnetic self-interactions.
- **Operator-Learning variants (AP-CON, GRTE-APNN)** [**Wu2025**, 5]: Integrating operator theories, we constructed variants like AP-CON for Gray radiative transfer using convolutional DeepONets. These advanced pipelines generalize across 10^4 distinct optical configurations seamlessly, reaching classical S_N solver accuracy but operating $10^3\times$ faster in inference.
- **Recurrent Temporal Extensions (RT-APNN)** [**RTAPNN2025**]: For pure time-dependent analysis, we developed Recurrent Temporal APNNs, employing GRU mechanisms with physical priors to capture long-time dynamics smoothly without suffering instability across varied timescales.

2.3 Asymptotic-Preserving Random Feature Methods

Beyond neural networks, in collaboration with Jingrun Chen (USTC), we developed an *asymptotic-preserving random feature method* (AP-RFM) [6] for multiscale radiative transfer. Random feature methods expand the solution in a random basis $\sigma(\omega_j \cdot x + b_j)$ with fixed random weights, offering mesh-free approximation with theoretically controllable error at $O(N)$ complexity. By encoding the micro–macro decomposition into both the basis construction and the least-squares fitting, AP-RFM achieves uniform accuracy across all Knudsen numbers at a cost substantially lower than neural network training. This result demonstrates that the AP design principle is a general mathematical framework—not a property specific to any particular machine learning architecture.

3 Fast Spectral Algorithms for Kinetic Equations

Why it matters. Classical fast algorithms play an indispensable dual role in my research program. They are the reference solvers that generate ground-truth data for training and validating all machine learning models, and they are independent scientific contributions that extend the computational frontier of kinetic theory. Without reliable, efficient classical solvers, the benchmarking of any AI-based method is scientifically unsound.

The direct evaluation of the Boltzmann collision integral at velocity-space resolution N^3 requires $O(N^6)$ operations per time step—completely intractable for three-dimensional non-equilibrium gas dynamics even on modern supercomputers. Fast spectral methods that exploit the convolution structure of the collision operator in Fourier space reduce this to $O(N^4 \log N)$ for the elastic Boltzmann operator, but the *inelastic* Boltzmann operator—governing granular gases, vibrationally excited molecular gases, and certain plasma-neutral interactions—had not been treated by any algorithm achieving better than $O(N^6)$ prior to our work.

During my postdoctoral appointment at Purdue University (mentored by Jingwei Hu), I developed a *fast spectral method for the inelastic Boltzmann collision operator* [7], achieving $O(N^2 \log N)$ complexity per velocity mode by uncovering a previously unrecognized convolution structure in the inelastic collision kernel after a suitable change of variables in Fourier space. This is, to our knowledge, the first sub-cubic algorithm for the inelastic operator. The method handles the full non-equilibrium collision dynamics and has been validated against Direct Simulation Monte Carlo (DSMC) benchmarks for heated granular gas flows across a broad range of inelasticity parameters.

The implementation is released as open-source software:

- `mazhengcn/fsm-inelastic-boltzmann`: the complete spectral solver for the inelastic Boltzmann operator.
- `mazhengcn/kipack`: a high-performance Python package supporting both elastic and inelastic Boltzmann operators, with GPU acceleration via JAX/CuPy and a clean NumPy interface.

These tools are actively used in our group and by external collaborators to generate training data for APNNs and DeepRTE, closing the loop between fast classical algorithms and modern machine learning.

Earlier work [8, 9] developed rigorous AP–UQ methods for uncertain transport and hyperbolic equations via stochastic Galerkin discretizations, establishing uniform spectral convergence from the kinetic regime to the diffusion limit in the presence of random inputs. These results provided both the mathematical framework and the design intuition that informed the APNN architecture, and they remain the most complete rigorous analysis of uncertainty quantification for multiscale kinetic theory.

4 Future Research Directions

My ongoing and planned research evolves along three interconnected frontiers, each combining fundamental mathematical questions with high-stakes engineering applications.

Foundation models for plasma and fusion confinement. Building on the DeepRTE paradigm, I am developing operator-learning foundation models for the Vlasov–Maxwell system governing fusion plasma dynamics. The scientific target is a surrogate model that replaces PIC simulations for tokamak transport calculations, accurate across the full kinetic-to-fluid spectrum and evaluable in milliseconds for real-time feedback control of plasma shape and stability—a prerequisite for commercially viable fusion energy. This project is being developed in collaboration with plasma physicists at SJTU's Institute of Natural Sciences. A key open challenge is the preservation of physical invariants (total energy, L^2 norm of the distribution function, entropy inequality) within the operator-learning architecture, connecting directly to the structure-preserving research thread below.

Convergence theory for generative models in inverse problems. Current diffusion-model-based inversion methods, including ODE-DPS, rely on heuristic justifications for the posterior score approximation step. I plan to develop a rigorous error analysis that bounds the Wasserstein distance between the learned approximate posterior and the true Bayesian posterior as a function of (i) the score approximation error, (ii) the ODE discretization step size, and (iii) the forward model accuracy. This connects F-principle theory—which governs the spectral convergence of the score network—with classical Bayesian regularization theory, providing what would be, to our knowledge, the first end-to-end convergence guarantee for generative inversion methods.

Structure-preserving operator learning. Inspired by symplectic integrators and AP schemes in classical numerical analysis, I aim to build operator-learning architectures that provably preserve nonlinear conservation laws (L^p mass, energy, entropy inequality) for Hamiltonian and dissipative PDE systems. The central technical challenge is to enforce these constraints as hard architectural invariants rather than loss-function penalties, since

penalty-based approaches are known to degrade in the stiff multiscale limit—precisely the regime of greatest scientific interest. Success here would generalize the APNN design principle to a broad class of physical constraints applicable to fluid mechanics, plasma physics, and molecular dynamics simulation.

5 Conclusion

My research develops a principled, mathematically rigorous approach to AI for Science that spans from foundational theory to production engineering. The unifying insight is that physical structure—the asymptotic limits, conservation laws, and variational principles that underlie kinetic equations and their inverses—can and must be encoded as hard constraints in the design of neural architectures, rather than as optional regularization. This produces methods that are not merely empirically successful but provably correct in the physically relevant limiting regimes. Equally importantly, every major methodological contribution in my group is accompanied by an open-source implementation that substantially lowers the barrier to adoption for scientists and engineers beyond the computational mathematics community. I intend to continue building this program at Shanghai Jiao Tong University, deepening international collaborations, and training computational mathematicians who can operate fluently at the interface of rigorous analysis and cutting-edge machine learning.

Selected References

[1] Y. Zhu, M. Tang, and Z. Ma. "DeepRTE: Pre-trained Attention-based Neural Network for Radiative Transfer". In: *Computer Methods in Applied Mechanics and Engineering* (2026).

[2] S. Jin, Z. Ma, and K. Wu. "Asymptotic-Preserving Neural Networks for Multiscale Time-Dependent Linear Transport Equations". In: *Journal of Scientific Computing* 94.3 (2023), p. 57.

[3] S. Jin, Z. Ma, and K. Wu. "Asymptotic-Preserving Neural Networks for Multiscale Kinetic Equations". In: *Communications in Computational Physics* 35.3 (2024), pp. 693–723.

[4] S. Jin, Z. Ma, and T.-a. Zhang. "Asymptotic-Preserving Neural Networks for Multiscale Vlasov–Poisson–Fokker–Planck System in the High-Field Regime". In: *Journal of Scientific Computing* 99.3 (2024), p. 61.

[5] K. Wu, X.-B. Yan, S. Jin, and Z. Ma. "Capturing the diffusive behavior of the multiscale linear transport equations by Asymptotic-Preserving Convolutional DeepONets". In: *Computer Methods in Applied Mechanics and Engineering* 418 (2024), p. 116531.

[6] J. Chen, Z. Ma, and K. Wu. "A Micro-Macro Decomposition-Based Asymptotic-Preserving Random Feature Method for Multiscale Radiative Transfer Equations". In: *Journal of Computational Physics* (2025).

[7] J. Hu and Z. Ma. "A Fast Spectral Method for the Inelastic Boltzmann Collision Operator and Application to Heated Granular Gases". In: *Journal of Computational Physics* 385 (2019), pp. 119–134.

[8] S. Jin, J.-G. Liu, and Z. Ma. "Uniform Spectral Convergence of the Stochastic Galerkin Method for the Linear Transport Equations with Random Inputs in Diffusive Regime and a Micro-Macro Decomposition-Based Asymptotic-Preserving Method". In: *Research in the Mathematical Sciences* 4.1 (2017), p. 15.

[9] S. Jin and Z. Ma. "The Discrete Stochastic Galerkin Method for Hyperbolic Equations with Non-smooth and Random Coefficients". In: *Journal of Scientific Computing* 74.1 (2018), pp. 97–121.